



Customer Churn Analysis & Prediction Dashboard



Project Overview

Customer retention is one of the most critical factors affecting business growth and profitability. This project presents a comprehensive **Customer Churn Analysis & Prediction Dashboard** developed using **Power BI** to analyze customer behavior, identify churn patterns, and provide actionable insights for improving customer retention.

The dashboard enables stakeholders to monitor churn-related KPIs, understand customer segments at risk, evaluate revenue impact, and make data-driven decisions to reduce customer attrition.



Business Objective

The primary objective of this project is to:

- Identify customers likely to churn.
 - Analyze factors contributing to customer attrition.
 - Measure customer retention performance.
 - Quantify revenue impact due to churn.
 - Support proactive customer retention strategies.
 - Enable data-driven decision-making through interactive visualizations.
-



Problem Statement

Customer churn can significantly impact a company's revenue and long-term growth. Acquiring new customers is often more expensive than retaining existing ones. Therefore, businesses need an effective analytical solution to understand customer behavior and detect churn risks before customers leave.

This dashboard helps organizations answer critical questions such as:

- Why are customers leaving?
 - Which customer segments have the highest churn rates?
 - What is the financial impact of customer churn?
 - Which customers should be targeted for retention campaigns?
-



Tools & Technologies Used

- Microsoft Power BI
- Power Query
- DAX (Data Analysis Expressions)

- Data Modeling
 - Data Transformation
 - Business Intelligence
 - Data Visualization
-

Project Workflow

1. Requirement Gathering

Understanding business goals, customer retention challenges, and stakeholder expectations.

2. Data Collection

Collecting customer demographic, transactional, subscription, and behavioral data.

3. Data Cleaning

- Handling missing values
- Removing duplicates
- Standardizing formats
- Data quality validation

4. Data Modeling

Creating relationships between multiple datasets to support efficient reporting and analysis.

5. Data Transformation

Using Power Query to prepare and transform raw data into an analysis-ready format.

6. DAX Calculations

Creating custom measures and KPIs for churn analysis.

7. Dashboard Development

Building interactive visualizations and executive-level reporting dashboards.

8. Insight Generation

Extracting meaningful business insights and recommendations.

Key Performance Indicators (KPIs)

Customer Metrics

- Total Customers
- Active Customers

- Churned Customers
- New Customers



Retention Metrics

- Churn Rate (%)
- Retention Rate (%)
- Customer Lifetime Value (CLV)
- Average Customer Tenure



Revenue Metrics

- Total Revenue
- Revenue at Risk
- Lost Revenue Due to Churn



Customer Engagement Metrics

- Customer Satisfaction Score
- Product Usage Frequency
- Support Ticket Analysis



Dashboard Features

Executive Summary Dashboard

Provides an overall business overview with key KPIs and performance indicators.

Customer Segmentation Analysis

Breaks down customers by:

- Age Group
- Gender
- Region
- Subscription Type
- Customer Value

Churn Analysis Dashboard

Analyzes:

- Churn Trends
- Churn by Customer Segment
- Churn by Demographics
- Churn by Product Category

Revenue Impact Dashboard

Evaluates:

- Revenue Loss
- Revenue at Risk
- High-Value Customer Churn

Predictive Insights Dashboard

Identifies:

- High-Risk Customers
- Churn Probability
- Retention Opportunities

Business Insights Generated

Key Findings

- High churn rates were concentrated within specific customer segments.
- Customer tenure strongly influenced retention rates.
- Certain subscription plans showed higher churn probability.
- Revenue loss was significantly impacted by a small percentage of high-value customers.
- Customer engagement metrics showed a strong correlation with retention.

Recommendations

Customer Retention Strategies

- Launch personalized retention campaigns.
- Improve customer onboarding experience.
- Offer loyalty and reward programs.
- Increase customer engagement initiatives.
- Monitor high-risk customers proactively.
- Develop targeted marketing campaigns for vulnerable segments.

Dashboard Preview

Add screenshots of your Power BI dashboard here:

```
/dashboard-screenshots
├─ Executive Dashboard.png
├─ Churn Analysis.png
├─ Customer Segmentation.png
```

```
├─ Revenue Impact.png
└─ Predictive Insights.png
```

Repository Structure

```
Customer-Churn-Analysis/
├─ Dashboard/
│   └─ Customer Churn Analysis & Prediction Dashboard.pbix
├─ Screenshots/
│   ├── Executive Dashboard.png
│   ├── Churn Analysis.png
│   ├── Customer Segmentation.png
│   └─ Revenue Impact.png
├─ Documentation/
│   └─ Project Report.pdf
└─ README.md
```

Skills Demonstrated

- Data Analysis
 - Business Intelligence
 - Data Visualization
 - Dashboard Design
 - DAX Development
 - Data Modeling
 - Power Query
 - KPI Development
 - Business Storytelling
 - Customer Analytics
-

Project Outcome

This project successfully transformed raw customer data into actionable business insights, enabling organizations to identify churn risks, improve customer retention, minimize revenue loss, and make strategic business decisions using interactive Power BI dashboards.

Author

Devansh Tyagi

 tyagidev037@gmail.com

 Marketing & Sales Associate | Data Analytics Enthusiast | Power BI Developer

 LinkedIn: <https://linkedin.com/in/devansh-tyagi-3260a3362>

★ **If you found this project helpful, don't forget to Star the repository!**